

System need to do!!!

Bayesian Hypothesis test, Bayes factor and Information criteria

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5.1 Tests in Bayesian framework

5.1.1 Frequentist tests

1. Hypothesis:

$$H_0 : \mu = \mu_0 \quad H_A : \mu \neq \mu_0$$

2. Test statistics

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad F = \frac{MSR}{MSE}$$

3. Sampling distribution under H_0
4. Critical region or p-value

5.1.2 The Bayesian Approach (one-sided case)

- We just need the posterior probability that the parameter θ falls in H_0 and H_A
- Hypothesis

$$H_0 : \theta \leq c \quad H_A : \theta > c$$

- $\Theta_0 = (-\infty, c]$ and $\Theta_A = (c, +\infty)$.
- Under posterior distribution:

- **Example 1:** Let Y_i be the number of strikes in a sequence of strikes before the cessation of the series and $Y_i|\theta \sim \text{geometric}(\theta)$, for $i = 1, \dots, 11$. The pmf is

$$p(y) = \theta(1-\theta)^y, \quad y = 0, 1, 2, \dots$$

And it is known that $\sum_{i=1}^{11} y_i = 148$.

1. The prior distribution is $p(\theta) \propto \sqrt{I(\theta)}$

2. Posterior of θ

3. $H_0 : \theta \leq 0.05, \quad H_A : \theta > 0.05$

4. Simulation

5. $H_0 : \theta = 0.1 \quad H_A : \theta \neq 0.1$

5.2 Bayes Factor

- Bayes Factor is used to compare models
- Compare two models M_1 and M_2 . Given data $\mathbf{y}$, consider the two models, M_1 and M_2 are not necessarily nested.

$$M_1 : p_1(\mathbf{y}|\boldsymbol{\theta}_1, M_1)$$

$$M_2 : p_2(\mathbf{y}|\boldsymbol{\theta}_2, M_2)$$

- Priors: $p(\boldsymbol{\theta}_1|M_1), p(M_1), p(\boldsymbol{\theta}_2|M_2), p(M_2)$

- **Bayes factor** is defined as

$$BF_{12} = \frac{p(\mathbf{y}|M_1)}{p(\mathbf{y}|M_2)}$$

- If $p(M_1) = p(M_2)$

- **Rule of Thumb for BF**

BF	Conclusion
$BF_{0A} \geq 1$	M_1 supported
$10^{-1/2} \leq BF_{0A} < 1$	Minimal evidence against M_1
$0.1 \leq BF_{0A} < 10^{-1/2}$	Substantial evidence against M_1
$0.01 \leq BF_{0A} < 0.1$	Strong evidence against M_1
$BF_{0A} < 0.01$	decisive evidence against M_1

- **Example 1:** $Y_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

- σ^2 is known, we want to test $H_0 : \mu = \mu_0$ $H_A : \mu = \mu_A$. Define BF_{0A}

- If σ^2 is known, but we want to test $H_0 : \mu = \mu_0$ $H_A : \mu \neq \mu_0$

- **Example 2:** 2003: HIV vaccine vs placebo given to 16395 individuals. 2006: HIV test

- 74 HIV positive reported in 8198 individuals receiving placebo
- 51 HIV positive reported in 8197 individuals receiving treatment
- Let p_1 and p_2 denote the probability of HIV positive in placebo and treatment population.

$$H_0 : p_1 = p_2 \quad H_A : p_1 \neq p_2$$

- **In frequentist framework**

- **Bayes Factor**

- **Some extensions**

- From Rouder et al (2009), Bayesian t tests for accepting and rejecting the null hypothesis, Psychonomic Bulletin & Review, 16(2), 225–237.
- One sample t -test: $Y_i \stackrel{iid}{\sim} N(\mu, \sigma^2), i = 1, \dots, n$
 - * $H_0 : \mu = \mu_0$ and $H_A : \mu \neq \mu_0$
 - * Defined $\delta = \frac{\mu - \mu_0}{\sigma}$, where the prior of δ is Cauchy distribution.
 - * And $\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$
 - * Consider $t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$
 - * The reference gave a form for the corresponding BF_{0A} , which is a function of t .
- Two sample t -test: $X_i \stackrel{iid}{\sim} N(\mu_1, \sigma^2), i = 1, \dots, n_1$ and $Y_j \stackrel{iid}{\sim} N(\mu_2, \sigma^2), j = 1, \dots, n_2$
 - * $H_0 : \mu_1 = \mu_2, H_A : \mu_1 \neq \mu_2$
 - * Priors: A Cauchy prior on $\delta = \frac{\mu_1 - \mu_2}{\sigma}$
 - * $\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$
 - * Consider

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}} / \sqrt{n^*}}, \text{ where } n^* = \frac{n_1 n_2}{n_1 + n_2}$$

- **Properties of BF**

- Generally we need proper priors of $[\theta_1|M_1]$ and $[\theta_2|M_2]$
- For simple hypothesis, it reduces to the likelihood ratios
- Unlike some other model comparison tools, it works for non-nested models